

BlinkIT Grocery Sales Analysis

Complete Power BI Project Analysis — Text-Only Documentation

This report documents the uploaded Power BI (.pbix) project in detail. No images, screenshots, or visual assets are included in this version. The analysis is based on the report layout, fields, measures, slicers, and visual configurations contained in the uploaded PBIX.

Project: BlinkIT Grocery Sales Analysis

Technology: Microsoft Power BI, DAX, Power Query, Data Modeling and Interactive Visualization

Report Page: Page 1

1. Executive Summary

The BlinkIT Grocery Sales Analysis dashboard is designed to provide a management-friendly view of grocery sales performance. The report combines headline KPIs with product, outlet, geographic-tier, outlet-size, outlet-establishment-year, and outlet-type analysis. Interactive slicers allow the user to filter the dashboard by outlet location type, outlet size, and item type.

The dashboard is particularly useful for answering questions such as: Which product groups generate the most sales? How does fat-content mix differ by outlet location? Which outlet sizes contribute most to sales? How has sales performance developed by outlet establishment year? Which outlet location tier and outlet type perform best?

2. Dashboard KPI Analysis

KPI	Dashboard-Level Value	Business Meaning
Total Sales	\$1.20M	Overall sales/revenue generated by the analyzed grocery records.
No. of Items	8,523	Number of item records represented by the dashboard KPI.
Average Sales	\$141	Average sales value at the row/item level used by the report.
Average Rating	3.9	Average rating associated with the analyzed items.

Note: KPI values are reported as dashboard-level values associated with the supplied project. A refresh in Power BI Desktop may recalculate values if the underlying model/source changes.

3. Data Model and Fields

The report references a primary table named **BlinkIT Grocery Data** and a supporting **Metrics** table. The primary table supplies product, outlet, sales, rating, visibility, location, size, and establishment-year fields. The Metrics table is used for metric selection in the dashboard.

Field	Role / Analysis
Sales	Primary sales/revenue measure used across charts and KPIs.
Item Fat Content	Categorizes products by fat-content group such as Low Fat and Regular.
Item Type	Product-category dimension used for category-level sales analysis.
Outlet Establishment Year	Time dimension used for outlet establishment trend analysis.
Outlet Location Type	Geographic market-tier dimension used for location analysis.
Outlet Size	Store-format dimension used to compare outlet-size contribution.
Outlet Type	Outlet-format dimension used in the management summary matrix.
Item Visibility	Measures/indicator used to evaluate product visibility in the outlet.
Rating	Rating field used to calculate average rating.

4. DAX Measure Analysis

The PBIX references the following core measures:

- **Total Sales** — SUM of the Sales column. Used as the main revenue KPI and across multiple charts.
- **Avg Sales** — AVERAGE of Sales. Used for average-sales KPI and outlet-type comparison.

- **No of Items** — COUNTROWS over the BlinkIT Grocery Data table. Used for item-count KPI and outlet-type comparison.
- **Avg Rating** — AVERAGE of Rating. Used for the average-rating KPI and outlet-type comparison.

The report also references a Metrics table and a Metrics/metric-selection mechanism. This allows a visual or slicer to switch the metric context rather than requiring separate static report pages.

5. Visual-by-Visual Analysis

5.1 Total Sales KPI Card

A card visual displays Total Sales. It provides the primary revenue headline for the dashboard and is intended for quick executive monitoring.

5.2 Average Sales KPI Card

A card visual displays Avg Sales. This helps place total sales in the context of average item-level sales performance.

5.3 Number of Items KPI Card

A card visual displays No of Items. It communicates the volume of item records contributing to the analysis.

5.4 Average Rating KPI Card

A card visual displays Avg Rating. This adds a quality/customer-perception indicator alongside financial KPIs.

5.5 Sales by Item Fat Content

A donut chart uses Item Fat Content as the category and Sales as the value. It is designed to compare the revenue contribution of Low Fat versus Regular products.

5.6 Fat Content by Outlet Location Type

A clustered bar chart compares Item Fat Content across Outlet Location Type. This supports a market-tier comparison of product mix and helps identify whether product preference differs between outlet location tiers.

5.7 Item Type Analysis

A bar chart uses Item Type as the category. It is intended to compare performance across grocery product categories and identify stronger or weaker item groups.

5.8 Sales by Outlet Establishment Year

An area chart plots Total Sales against Outlet Establishment Year. This creates a historical trend view for sales associated with outlet establishment periods.

5.9 Sales by Outlet Size

A donut chart compares Total Sales across Outlet Size categories. It supports evaluation of store-format contribution and helps management assess which outlet size is commercially strongest.

5.10 Sales by Outlet Location Type

A funnel visual compares sales across Outlet Location Type. It gives a compact ranking-style view of Tier 1, Tier 2, and Tier 3 market performance.

5.11 Outlet Type Summary Matrix

A pivot/matrix visual compares Outlet Type using Total Sales, No of Items, Avg Sales, Avg Rating, and Item Visibility. This is the most detailed management comparison in the report because it combines financial, volume, quality, and visibility indicators.

6. Filter and Slicer Analysis

- **Metrics** — Provides metric-selection functionality so users can change the analytical measure used by supported visuals.
- **Outlet Location Type** — Filters the dashboard by location tier/market type.
- **Outlet Size** — Filters the dashboard by outlet-size category.

- **Item Type** — Filters the dashboard by product category.

The filter panel is intentionally positioned as a dedicated dashboard control area. This makes the report suitable for interactive exploration during management reviews or portfolio demonstrations.

7. Business Questions Answered

- What is the overall sales performance?
- How many item records are represented in the analysis?
- What is the average sales value?
- What is the average rating?
- Which fat-content group contributes more sales?
- How does fat-content performance vary by outlet location tier?
- Which item types are stronger or weaker in sales?
- How does sales performance vary by outlet establishment year?
- Which outlet size contributes the greatest sales?
- Which outlet location type contributes the greatest sales?
- How do outlet types compare on sales, item count, average sales, rating, and visibility?

8. Business Insights Framework

- **Revenue performance** — Total Sales is the top-level indicator for evaluating overall commercial performance.
- **Product mix** — Fat-content and item-type analysis can guide assortment, pricing, promotion, and inventory decisions.
- **Geographic strategy** — Outlet Location Type analysis can reveal which market tiers deserve greater attention.
- **Store format** — Outlet Size and Outlet Type analysis can help identify formats that produce stronger sales or better overall performance.
- **Outlet maturity** — Establishment-year analysis can provide historical context for sales development.
- **Customer/product perception** — Average Rating provides a quality/perception dimension alongside revenue.
- **Visibility** — Item Visibility in the outlet summary can be considered when investigating category or outlet performance.

9. Recommended Analytical Interpretation

The dashboard should be interpreted as a combination of descriptive and comparative analytics. The KPI layer establishes the overall business baseline. The category and outlet charts then explain where sales are coming from. The slicers allow the analyst to test whether conclusions remain consistent when the data is narrowed to a specific market tier, outlet size, or product category.

For example, an analyst can select a specific Outlet Location Type and then review the KPI cards, fat-content chart, item-type chart, outlet-size distribution, and outlet-type matrix. This supports a drill-down style workflow from overall performance to a more focused business segment.

10. Data Analysis Workflow

1. Load the BlinkIT Grocery dataset into Power BI.
2. Inspect fields and establish the analytical dimensions and measures.
3. Use Power Query for required data preparation and type/format validation.
4. Create or use the sales, average-sales, item-count, and rating measures.
5. Build KPI cards for executive-level monitoring.
6. Build product-level visuals for fat content and item type.
7. Build outlet-level visuals for outlet size, location type, and outlet type.
8. Build establishment-year trend analysis.
9. Add slicers for interactive segmentation.
10. Validate cross-filtering and visual consistency before publishing.

11. Data Quality and Validation Checklist

- Verify Sales is numeric and contains valid values before aggregation.
- Verify Rating is numeric before calculating Avg Rating.
- Verify Outlet Establishment Year is treated as a year/category suitable for trend analysis.
- Standardize category labels for Item Fat Content, Item Type, Outlet Size, Outlet Type, and Outlet Location Type.
- Check for blank or unexpected category values.
- Check duplicate records where duplication would distort item counts or sales.
- Validate Total Sales against the source dataset after refresh.
- Validate that slicers cross-filter the intended visuals.
- Check that KPI values change correctly under filters.

12. Strengths of the Power BI Project

- Clear KPI-first dashboard structure.
- Multiple dimensions for product and outlet analysis.
- Interactive slicers for focused analysis.
- Combination of financial, volume, rating, and visibility metrics.
- Historical outlet-establishment trend.
- Management-friendly matrix for outlet-type comparison.
- Use of DAX measures rather than relying only on raw columns.

- Suitable for a Power BI portfolio project because it demonstrates both visualization and business-analysis thinking.

13. Suggested Improvements

- Add a dedicated Top 5 / Bottom 5 Item Type analysis.
- Add sales growth or year-over-year analysis if a suitable time basis is available.
- Add conditional formatting to the outlet summary matrix.
- Add tooltip pages for outlet and item-level details.
- Add a dedicated Item Visibility KPI if visibility optimization is a key business goal.
- Add a decomposition-tree or drill-through page for root-cause analysis.
- Add a clear data-refresh date and source note to improve dashboard governance.

14. Portfolio / GitHub Project Description

BlinkIT Grocery Sales Analysis | Power BI

Developed an interactive Power BI dashboard to analyze grocery sales performance across product categories, fat content, outlet types, outlet sizes, location tiers, and establishment years. Created DAX-based KPIs for Total Sales, Average Sales, Number of Items, and Average Rating, and implemented interactive slicers for business-focused analysis and reporting.

15. Resume Bullet Points

- Built an interactive Power BI dashboard to analyze grocery sales across item types, outlet types, outlet sizes, and location tiers.
- Developed DAX KPIs for Total Sales, Average Sales, Number of Items, and Average Rating.
- Created visual analysis for product fat content, category performance, outlet establishment trends, outlet size, and market-tier sales.
- Implemented interactive slicers and an outlet summary matrix to support business-focused segmentation and decision-making.

16. Technical Skills Demonstrated

Area	Skills Demonstrated
BI Tool	Microsoft Power BI Desktop
Data Preparation	Power Query / data transformation and validation
Analytics	Descriptive, comparative, category, outlet, and trend analysis
DAX	SUM, AVERAGE, COUNTROWS and metric-selection logic
Visualization	Cards, donut charts, bar charts, area chart, funnel, matrix, slicers
Data Modeling	Primary data table plus supporting Metrics table
Business Intelligence	KPI monitoring, segmentation, outlet comparison and decision support

17. Final Conclusion

The BlinkIT Grocery Sales Analysis Power BI project provides a complete dashboard-based approach to understanding grocery sales performance. It combines high-level KPIs with product, outlet, geographic-tier, size, and establishment-year analysis. The interactive filter design allows users to move from an overall business view to focused segment analysis.

From a portfolio perspective, the project demonstrates practical Power BI skills: data modeling, DAX measures, KPI creation, interactive filtering, multiple visualization types, and conversion of raw business data into decision-oriented insights. The strongest next step would be to extend the dashboard with growth analysis, ranking views, drill-through pages, and stronger conditional formatting.

End of report — Text-only version. No images or screenshots are included.